Intro Microelectronic Circuits with Laboratory

BJT Collector-base Coupled Non-Inverting Schmitt Trigger Lab

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1. Introduction

Schmitt triggers are used in a variety of applications from trigger inputs for microcontrollers to crude 1-bit DACs to oscillators. They can be built from a variety of components; namely, MOSFETs, BJTs, and discrete Operational Amplifiers. They work because of hysteresis, the output not being directly related to the input but rather dependent on the state of the system perviously, which allows the Schmitt trigger to output a high or low voltage based on two different threshold voltages.

2. Circuit Description

The specific Schmitt trigger circuit that we will be using is called the BJT Collector-base Coupled Non-Inverting Schmitt Trigger. It is comprised of 5 resistors and 2 BJTs in a configuration that resembles a SRAM cell. The output voltage will be the voltage at the collector of Q2. The circuit diagram below shows how this circuit is built. Note that for simplicity, we will assume that R1 is equal to R2 and R3 is equal to R4. The values of the resistors are as follows: R1 and R2 are 1.1 kΩ, R3 and R4 are 49.9 kΩ, and R5 is 4.9 kΩ. The BJTs used are the 2N23904 BJTs.

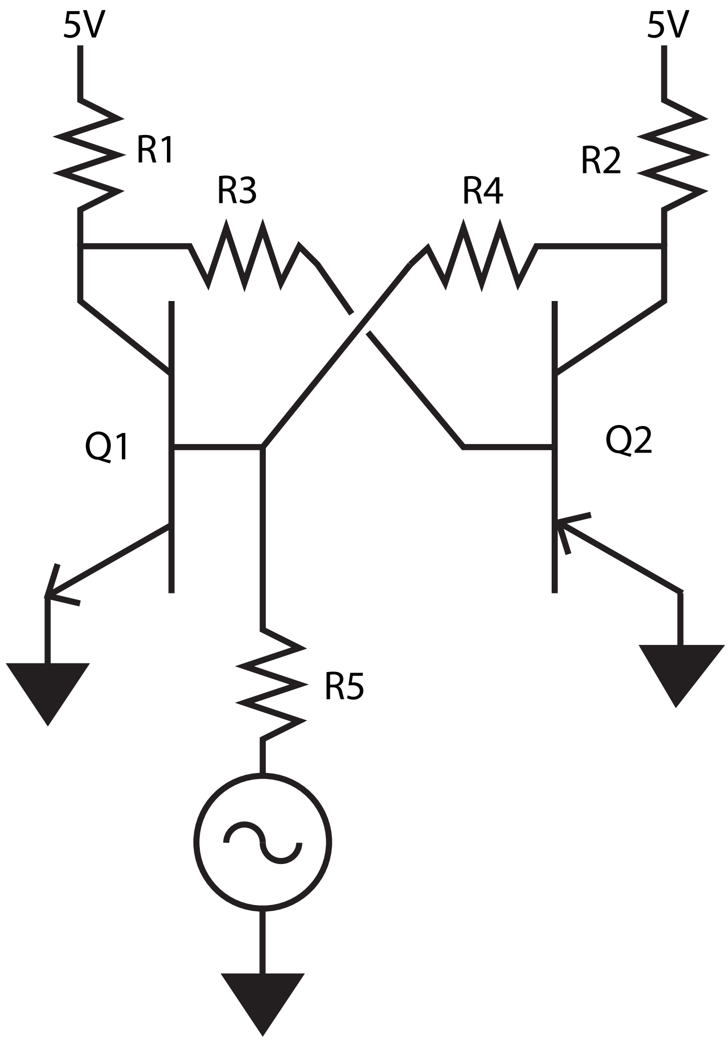

Figure 1. BJT Collector-base Coupled Non-Inverting Schmitt Trigger Circuit.

3. INCOMPLETE - Circuit Analysis

Before building this circuit, we used LTSpice to simulate the circuit which allowed us to easily change the resistance values of in the circuit to see how the circuit would respond to a sine wave input. We observed the following:

- The Output High Voltage is determined by the voltage divider ratio between R2 and R4 where they are a voltage divider.
- There has to be a gap between the R1, R2 values and the R3, R4 values; R1, R2 must be smaller than R3, R4.
- When Q1 or Q2 turns on, it sucks current from the other base of other BJT and turns the other BJT off at the same time.

4. Experiment 1: Schmitt Trigger Voltage Characterization

4.1. Experimental Setup

In this experiment, we set up the circuit as shown in Figure 1 and measured the output voltage of the circuit. We swept the voltage from V_{ee} (GND) to V_{cc} and then back to V_{ee}. We are characterizing the Upper and Lower Threshold Voltages of the circuit (represented by V_{UT} and V_{LT} respectively) and then calculating the Hysteresis Voltage (represented by V_H) by subtracting the Upper Threshold Voltage from the Lower Threshold Voltage.

$$V_H = V_{UT} - V_{LT}$$

With the same data, we also extracted the Output High and Low Voltage (represented by V_{OH} and V_{OL} respectively) by looking at what the average high voltage was and what the average low voltage was. Afterwards, we plotted the input voltage vs. the output voltage to get a hysteresis curve which shows us the output high and low voltages along with the low and high threshold voltages. This plot fully characterizes the voltage characteristics of the Schmitt trigger.

Experiment Deliverables: Hysteresis Curve Plot; Table of Upper Threshold, Lower Threshold, Output High, Output Low, and Hysteresis Voltages; Plot of Output Voltage on top of the Input Voltage.

4.2. Results

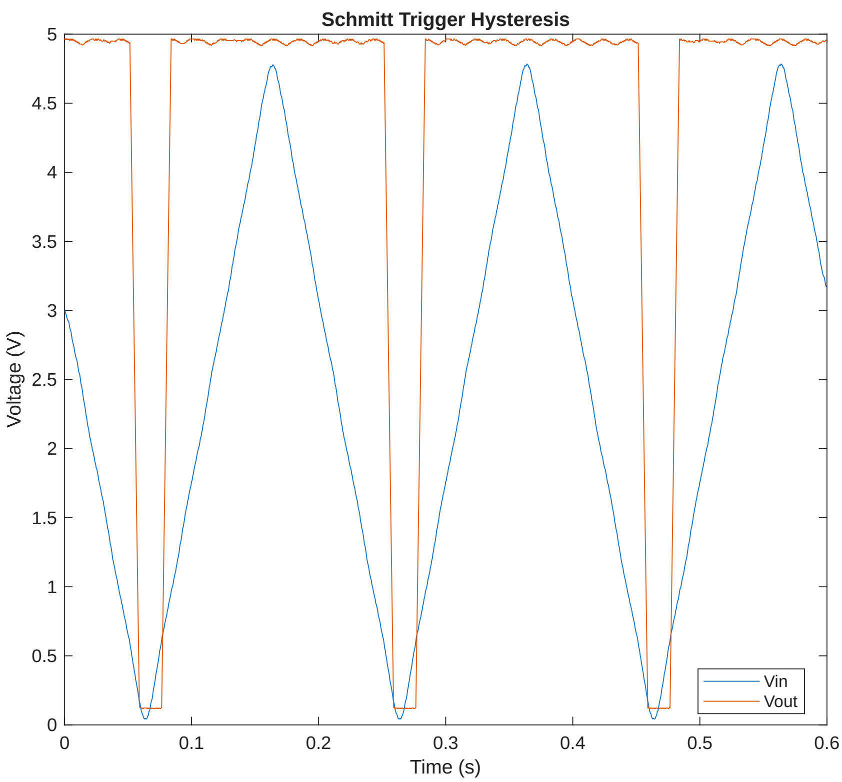

Figure 2. Voltage behavior of Schmitt Trigger. Notice how the circuit transitions to output low at a lower threshold than when it transitions to output high.

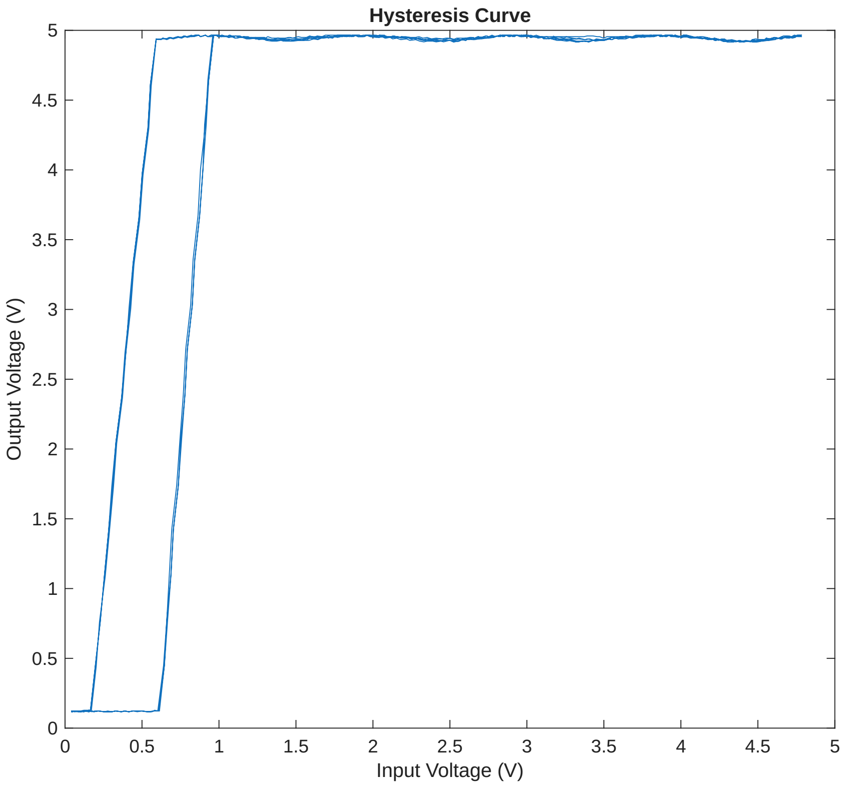

Figure 3. Hysteresis Curve of Schmitt Trigger.

Upper Threshold Voltage, V_{UT}	Lower Threshold Voltage, V_{LT}	Output High Voltage, V_{OH}	Output Low Voltage, V_{OL}
0.843 V	0.421 V	4.92 V	0.123 V

Hysteresis Voltage, V_H
0.421 V

Table 1. Schmitt Trigger Voltage Characteristics Values.

4.3. Analysis

We inputted a 5 Hz, 50 % duty cycle triangle wave into the circuit. From the data we collected, we were able to see how there are two different thresholds at which the output voltage changes from high to low or vice versa. This validates that there is hysteresis in the circuit which is also seen in the hysteresis curve plot.

4.4. Discussion

This experiment shows how the Schmitt trigger works by having two different thresholds for its output voltages. This particular circuit essentially has a rail-to-rail output.

5. Experiment 2: Schmitt Trigger Frequency Response Characterization

5.1. Experimental Setup

In this experiment, we used the circuit as shown in Figure 1 and inputted a 2.5 V_{DC} offset, 50% duty cycle sine wave at 5 Hz. We measured the slew rate by measuring the slope of the rising edge of the signal on the output of the circuit from 10% of the output voltage to 90% of the output voltage. This value represents how fast the circuit can drive a change in voltage.

$$SR = \max\left(\frac{d}{dt}[V_{out}]\right)$$

The output swing was measured by getting the average output high value and subtracting it from the average output low value. This value tells us how much the output can vary in voltage.

$$V_{OS} = V_{OH} - V_{OL}$$

The rise time and fall time were calculated by measuring the time between the start of the rise of the output voltage to the end of that rise. The same procedures were followed to measure fall time. The propagation delay is calculated by determining the time at which the input waveform reaches the threshold voltage of the circuit and subtracting that time from the time when the output starts to rise.

Experiment Deliverables: Plots showing the output Propagation Delay, Rise Time, Fall Time, Slew Rate, and Output Swing for the given input signal.

5.2. Results

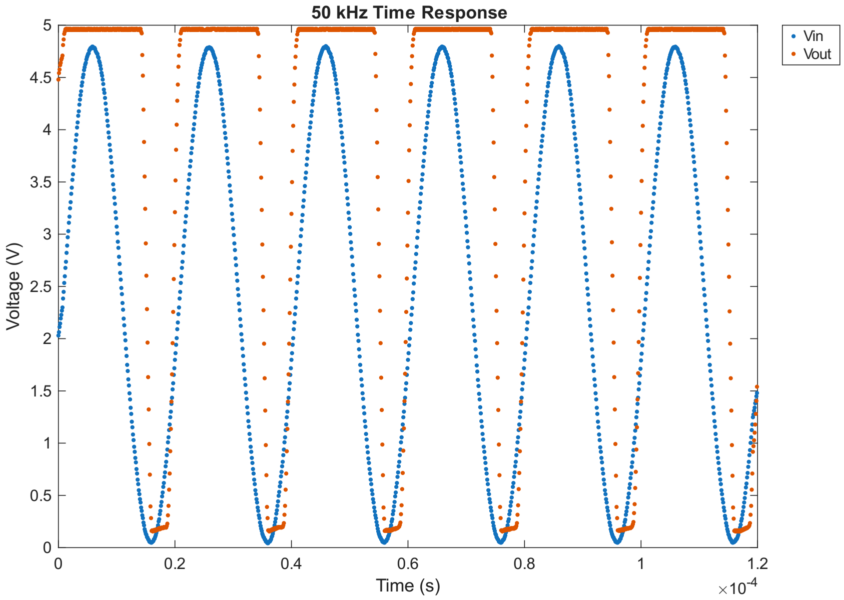

Figure 4. Voltage behavior of Schmitt Trigger. Notice how the circuit transitions to output low at a lower threshold than when it transitions to output high.

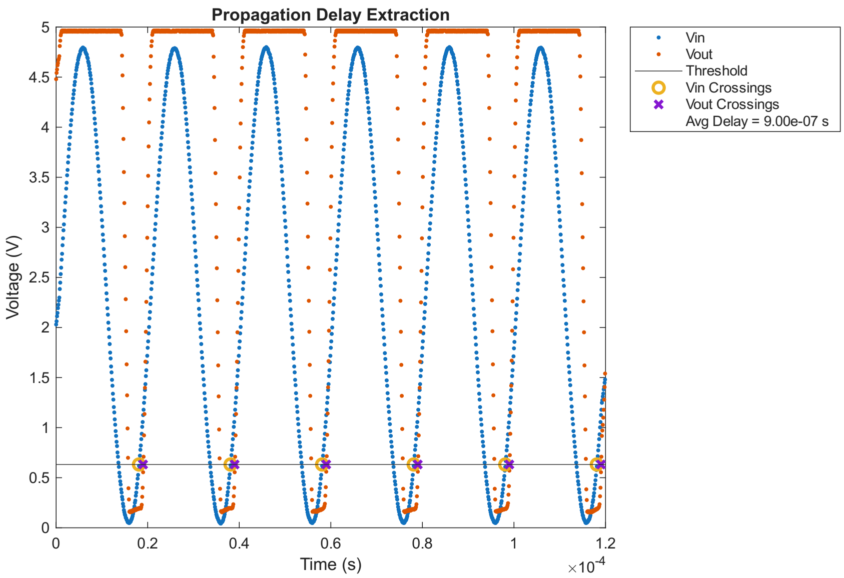

Figure 5. Voltage behavior of Schmitt Trigger. Notice how the circuit transitions to output low at a lower threshold than when it transitions to output high.

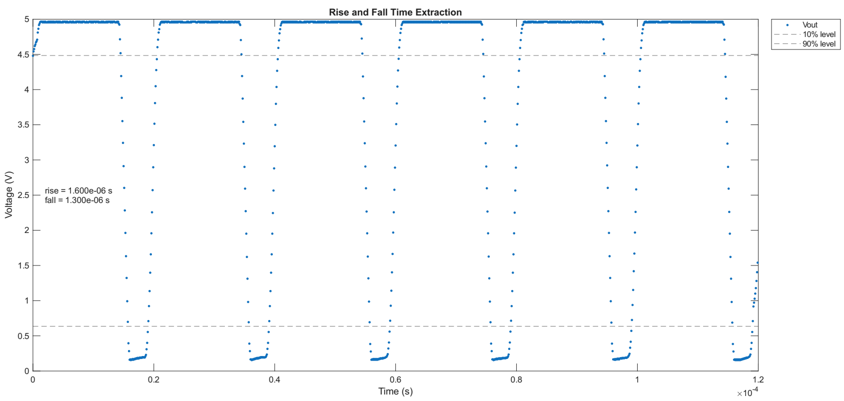

Figure 6. Voltage behavior of Schmitt Trigger. Notice how the circuit transitions to output low at a lower threshold than when it transitions to output high.

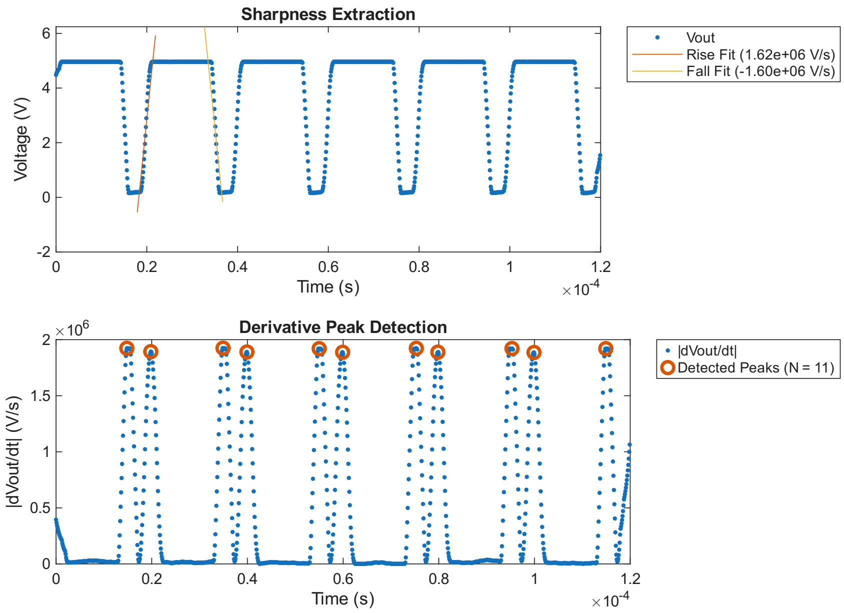

Figure 7. Voltage behavior of Schmitt Trigger. Notice how the circuit transitions to output low at a lower threshold than when it transitions to output high.

5.3. Analysis

5.4. Discussion

6. Experiment 3: Schmitt Trigger Frequency Response Trends

6.1. Experimental Setup

In this experiment, we used the circuit as shown in Figure 1 and inputted a 2.5 V_{DC} offset, 50% duty cycle sine wave at various frequencies between 5 Hz and 200 kHz. We wanted to see how the behavior of the circuit would change based on the input waveform frequency by measuring the propagation delay, rise time, fall time, slew rate, and output swing. The propagation delay describes how long it takes for the circuit to respond to the input. The slew rate describes how long it takes the circuit to drive the output. The output swing describes the range of voltage the circuit can drive on the output side. The switching sharpness describes how clear the transition was between high and low.

Experiment Deliverables: Plots showing Hysteresis Curve/Loop, Propagation Delay, Switching Sharpness, Slew Rate, and Output Swing varying with Frequency; Plots of input and output waveforms at a low, medium, and high frequency.

6.2. INCOMPLETE - Results

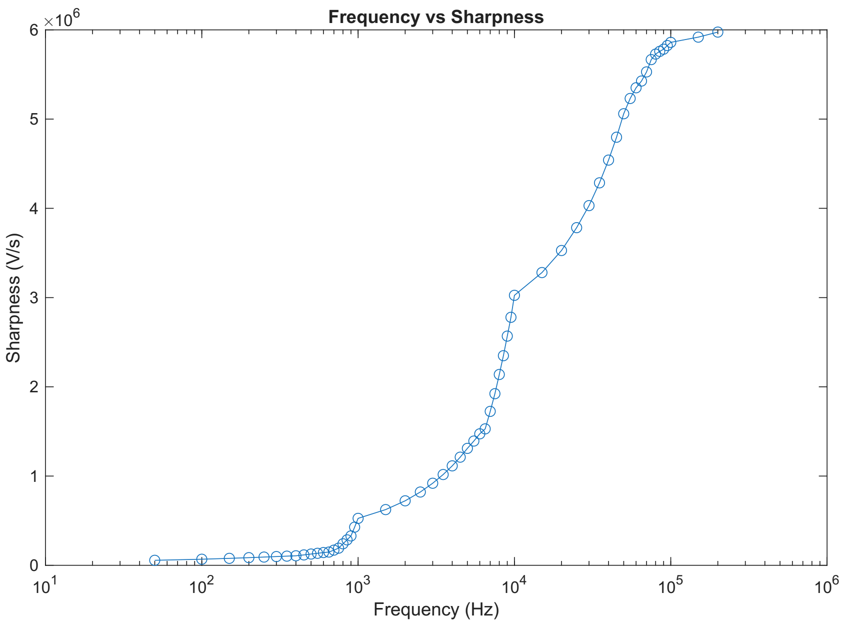

Figure 8. Slew rate relationship with frequency.

This plot shows how the slew rate changes with frequency. The slew rate decreases because the circuit has less time to drive up or down the circuit for a given period the higher the frequency becomes.

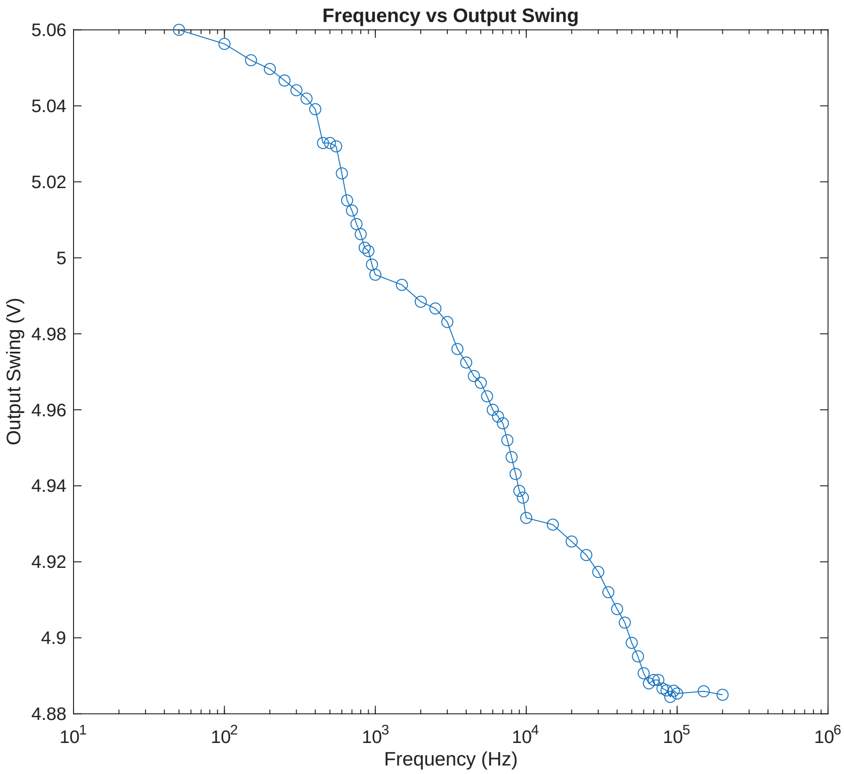

Figure 9. Output swing relationship with frequency.

This plot shows how the output swing decreases with frequency. This is because at higher frequencies, the circuit does not have enough time to drive the output high enough to it's theoretical output high voltage or low enough to its theoretical output low voltage.

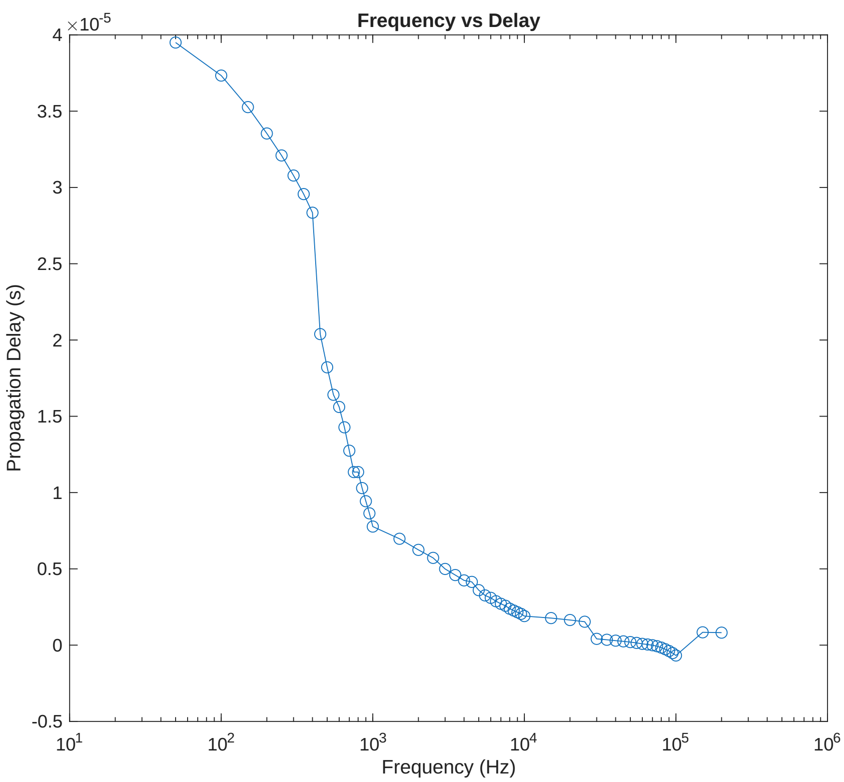

Figure 10. Propagation delay relationship with frequency.

This plot shows how the propagation delay increases as frequency increases.???

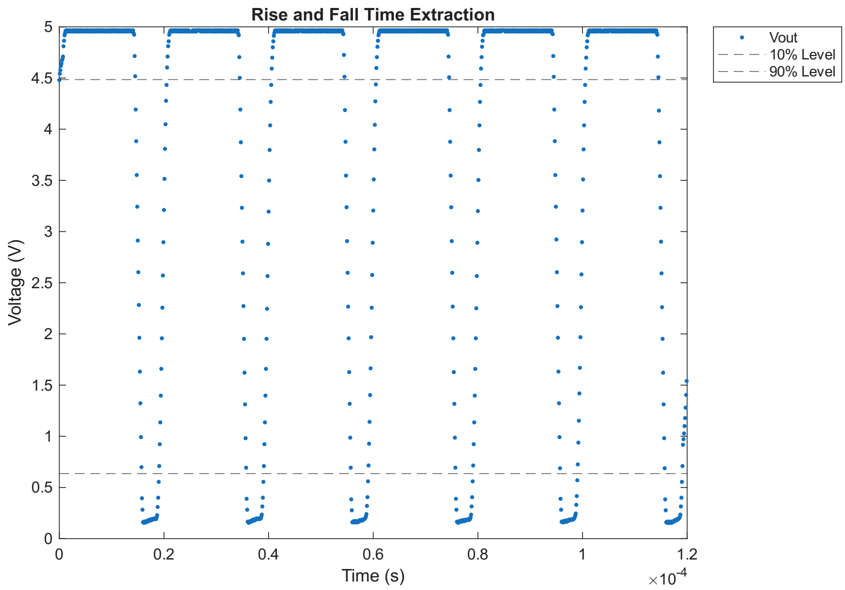

Figure 11. Propagation delay relationship with frequency.

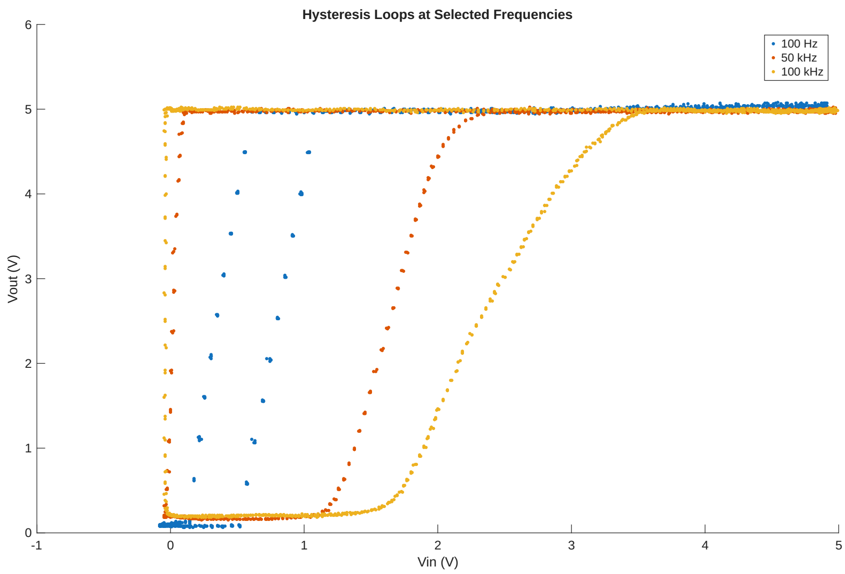

Figure 12. Hysteresis loop relationship with frequency.

As the frequency increases, the hysteresis loop becomes less and less sharp and becomes wider. This is because of how the slew rate and propagation delay change as frequency increases. As the slew rate decreases, the less steep the slope of the change from output high to output low and vice versa are. The larger the propagation delay, the wider the hysteresis loop becomes. The switching points shift due to time delay, because the input continues changing during the delay.

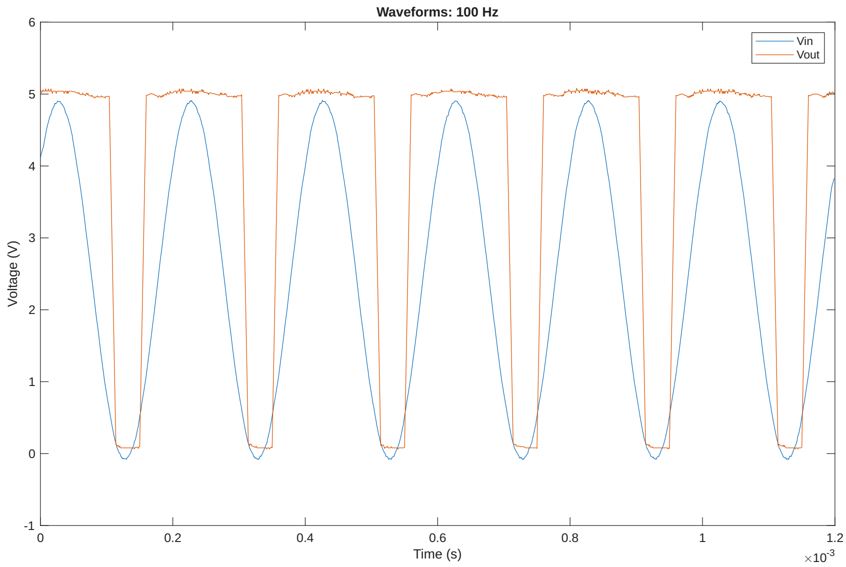

Figure 13. Low frequency waveform input output relationship.

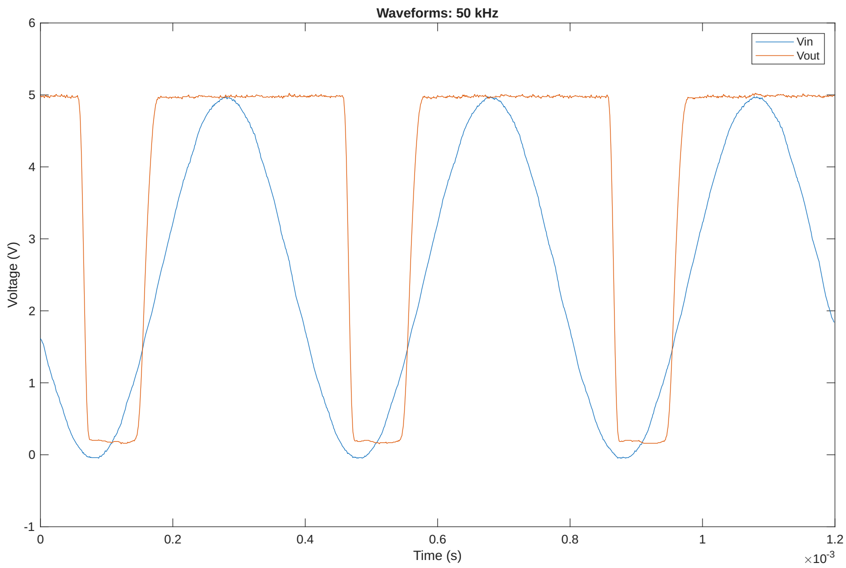

Figure 14. Medium frequency waveform input output relationship.

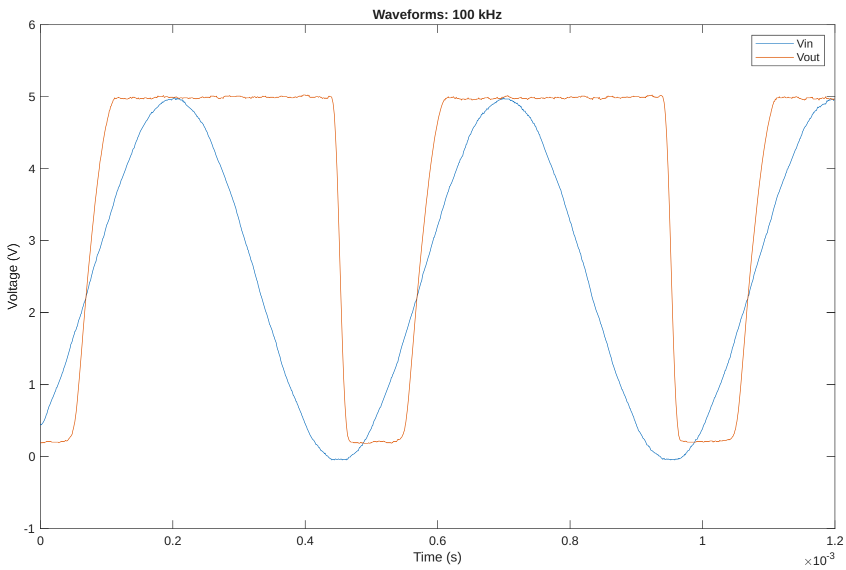

Figure 15. High frequency waveform input output relationship.

6.3. Analysis

As the frequency increases past 10 kHz, the output of the circuit starts to struggle to reach the output high and low values. By 100 kHz, the slowness of the change in output voltage becomes more noticeable. This helps explain why Schmitt triggers are rated for certain speeds, at very high speeds they stop working as expected.

6.4. Discussion

In most of the figures (specifically Figures 8 through 12), we observed jagged corners. We think this may be because of the sampling rate of the oscilloscope used when measuring the voltage at the input and output of the circuit because it was changed about every decade around the frequency where we observe the corner. This is because the sampling rate determines the resolution of the samples (and how far apart they are in time) which would affect the rates calculated using those time increments. In addition, these jumps occur around the same frequencies across all of the plots which points towards this being a measurement error.

7. Summary

In this lab, we learned a lot about how the characteristics of the Schmitt trigger change with the selection of resistors used and input waveform frequency. The particular Schmitt trigger that we are using is usable until about 200 kHz but starts showing issues with its output around 100 kHz. We also learned that the particular resistors that we used affected the output high and low voltages a lot.

8. Sources

- Schmitt Trigger Article - https://en.wikipedia.org/wiki/Schmitt_trigger
- Common Collector Amplifier Article - https://www.electronics-tutorials.ws/amplifier/common-collector-amplifier.html
- Schmitt Trigger Explained (Design of Inverting and Non-inverting Schmitt Trigger using Op-Amp) Video - https://youtu.be/5-ohKRWeod4?si=8VM2gfX6_NjWiwgg